

EdulcoWater Smart Sens Wi-Fi

Smart Sens model Ph-Orp for Pools/Aquarium:

() pH; () ORP; () Temperature; () 4 Timers

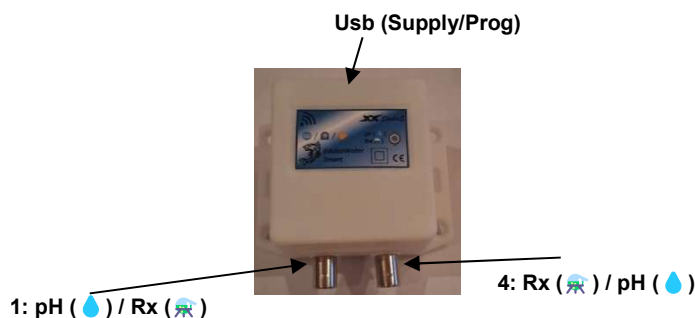


Fig.1a General View Smart Sens Ph-Orp

Smart Sens model Temp-EC for Hydroponics / Solar / HVAC / Thermal Systems:

() EC; () TDS; () Salinity; () Temperature; () 4 Timers
() Temperatures x2; () Δ Differential Temperature;
() $\emptyset$ Average Temperature; () 4 Timers

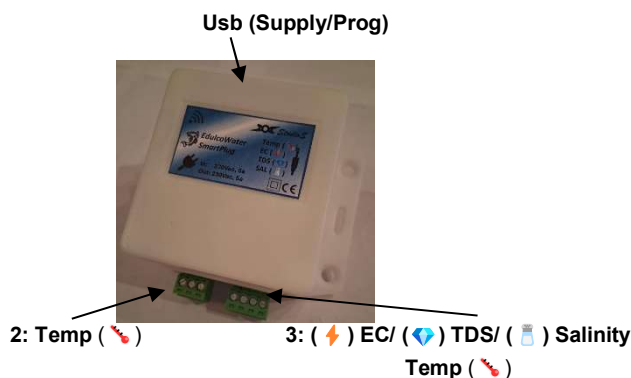


Fig.1b General View Smart Sens Ph-Orp

Product Description:

EdulcoWater Smart Sens is a family of two compact Wi-Fi monitoring and control modules designed for dedicated applications.

The **Smart Sens pH/ORP** model is intended for swimming pools, aquariums, and general water treatment applications, while the **Smart Sens EC/Temperature** model is designed for hydroponics, solar thermal systems, HVAC, and temperature monitoring applications.

Unlike the EdulcoWater Industrial controllers, Smart Sens modules do not include a display or onboard relay outputs. They operate as compact Wi-Fi controllers that can be accessed directly through their integrated web interface or seamlessly integrated into Smart Home and industrial automation platforms.

In automation systems such as Home Assistant, the Smart Sens modules provide real-time sensor measurements and

control logic, while relay activation can be managed by an external supervisory system or by dedicated EdulcoWater Smart Relay modules. This modular architecture provides maximum flexibility, allowing monitoring, decision-making, and power switching to be distributed according to the requirements of the application.

EdulcoWater Smart Sens Modules supports **5 different sensor technologies**:

Module Smart Sens Ph-Orp:

 pH

 ORP / Redox

Module Smart Sens EC-Temp:

 Digital temperature sensors (DS18B20)

 Analog NTC temperature

 EC /  TDS /  Salinity

Advanced  **thermal operating modes** allow intelligent multi-point temperature control for demanding applications such as solar thermal systems, HVAC automation, circulation pumps, heat exchangers, fireplaces with water jackets, and general industrial thermal processes.

Available modes include:

-Standard temperature control (single or dual sensor based)

-Dual temperature monitoring

-Differential temperature control (ideal for solar collector / tank ΔT applications)

-Average temperature control (combined thermal regulation based on the mean value of two measurement points)

The Modules provide **2 fully independent control channels for each module**. Every channel is configurable according to the selected application. The following architecture is available:
Control Architecture

Module Smart Sens Ph-Orp:

Control 1:  pH or  ORP / Redox

Control 4:  pH or  ORP / Redox

Module Smart Sens EC-Temp:

Control 2:  digital temperature (DS18B20)

Control 3:

Conductivity Family:  EC or  TDS or  Salinity or

Temperature Family:  Analog temperature or  Δ

Differential temperature (Control 3 – Control 2) or Σ average temperature (Control 3 + Control 2)/2

Each control channel includes an independent **time modulation timer** ().

This allows cyclic ON/OFF modulation of the control output, useful for applications such as:

EdulcoWater Smart Sens modules are configured, monitored, and operated entirely through the integrated Wi-Fi web server. The user interface is accessible from any smartphone, tablet, or PC using a standard web browser, with no dedicated application required.

The systems include an integrated **Wi-Fi web server**, making the controller accessible from any smartphone, tablet, or PC using a standard web browser — **no app installation required**.

EdulcoWater supports multiple connection modes:

-**Wi-Fi Access Point mode** for direct local connection (**no internet required**)

-**Wi-Fi Station mode** for integration into your existing home or industrial network

-**Remote worldwide access** through router port forwarding

-**HTTP command interface** for smart home and automation integration

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-The device also supports industrial automation integration through the integrated Modbus TCP Server interface, allowing communication with **PLCs, HMIs, and SCADA systems**.

The integrated web platform provides a modern and convenient interface for:

- live monitoring of all control channels
- sensor calibration
- control configuration
- timer management
- Wi-Fi configuration
- remote diagnostics

Advanced intelligent cloud-connected services are also available:

NetSky AI Copilot

AI-assisted support for:

- troubleshooting technical problems
- answering product and manual-related questions
- guided diagnostics
- assisted configuration suggestions
- automation command generation

Sentinel Monitoring

Advanced alarm supervision with:

- minimum / maximum threshold monitoring
- persistence filtering to avoid false alarms
- automatic alarm notifications
- recovery notifications
- email alert messaging

Reporter Data Logging

Integrated reporting and historical monitoring with:

- trend visualization
- scheduled email reporting with Excell data

The Smart Sens modules are powered through a standard **5 V USB connection**, allowing them to be supplied by any common USB power adapter, computer USB port, or compatible power bank.

The same USB connection is also used for firmware programming and development. The modules are fully compatible with the **EducoWater by Fabiano** open-source Arduino library, enabling developers to create custom applications, extend the device functionality, and integrate the modules into their own automation projects.

The **EducoWater by Fabiano** library is freely available and includes examples and documentation to simplify software development and system integration.

Installation application example:

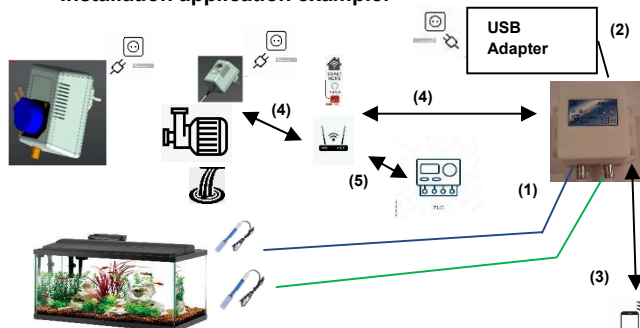


Fig.2 Typical Installation

1. Connect the Sensors

Connect the sensors according to the specific application:

-BNC Connector:

pH or **ORP / Redox probe**

-Terminal Block Connector:

DS18B20 digital temperature sensor

-Terminal Block Connector:

EC probe or

Analog temperature sensor

2. Power Connection

Power the Smart Sens module using a standard 5 V USB power supply, such as a common smartphone USB charger or any other regulated 5 V USB adapter.

After connecting the USB power supply, the device starts automatically and becomes available through its integrated Wi-Fi interface.

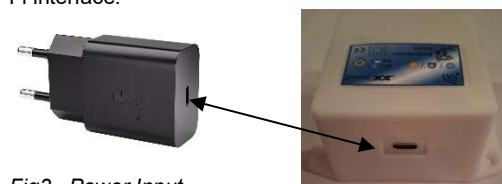


Fig3.. Power Input

3. Connect a Mobile Device

Use your smartphone or tablet to access the device's interface.

This can be done by connecting directly to the device's local hotspot or through your home Wi-Fi network.

4. Connect to Wi-Fi (Optional)

For remote monitoring and control, connect the device to your Wi-Fi network. Follow the on-screen instructions in the mobile interface to configure your network credentials.

When integrated with a Smart Home or industrial automation platform such as **Home Assistant**, the Smart Sens modules can be used to implement complete monitoring and control applications

The measured values and control status can be used to create custom automation rules that drive external actuators through **EducoWater Smart Relay** modules or other compatible relay devices. This distributed architecture provides a flexible solution for applications such as swimming pools, aquariums, hydroponics, solar thermal systems, HVAC installations, and general process automation.

5. PLC / SCADA Connection (Optional)

Connect the PLC, HMI, or SCADA system to the same network as the EducoWater device. Communication is available through the integrated **Modbus TCP Server** interface, allowing external controllers such as **Siemens S7-1200/S7-1500**, **Schneider Electric Modicon**, and other Modbus TCP compatible systems to read and write device registers.

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Sensors Wiring Smart Sens Ph-Orp:

Input (BNCs)	Description
Ph/Orp	Sensor 1
Orp/Ph	Sensor 4



Sensor 1

Sensor 4

Sensors Wiring Smart Sens EC-Temp:

Label	Pin	Description
Temperature	Vcc	V+
	DI	Digital Input
	0V	GND
Temperature 2/ EC-SAIL-TDS	Ec1	EC pin 1
	Ec2	EC pin 2
	T1	EC pin 3 / NTC pin 1
	T2	EC pin 4 / NTC pin 2



Sensor 2: Temp ()

Sensor 3: () EC/ () TDS/ ()
Salinity /Temp ()

Wi-Fi Connection: Direct Mode.

The system can connect to any device that supports Wi-Fi connectivity, such as smartphones, tablets, or PCs.

The device creates its own Wi-Fi access point. Simply connect your device to the EducoWater network to access the interface without requiring an internet connection.

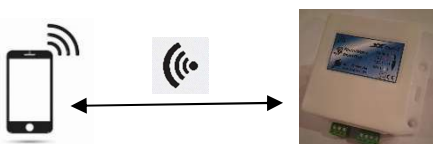
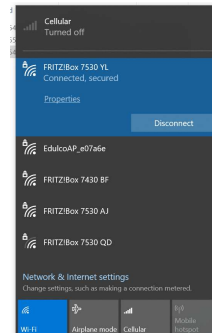
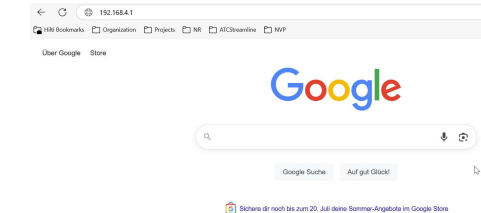


Fig.4 Direct mode WiFi connection

-Open the Wi-Fi settings on your smartphone, tablet, or PC and look for a Wi-Fi network named **EducoAP_XXXX** (the "XXXX" is a unique identifier for your device).



-Select the network and connect — no internet is required.
-Once connected, open your web browser (for an example Internet Explorer or Google) and enter the device's local IP address **192.168.4.1** to access the control interface.
-Default Password: "**123456789**"



Studios Web Page Interface

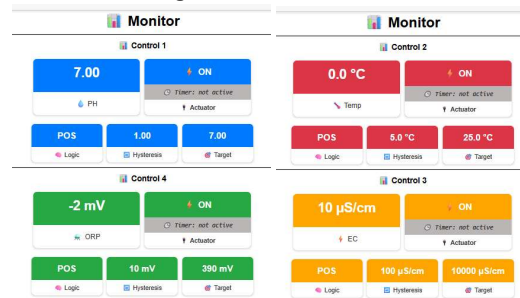


Fig.5 Web Page Interface

This is the main page that opens by default when accessing the device.

At the top (1), the navigation bar provides quick access to the following pages:

- **Monitor** – Shows system status.
- **Sensors** – Sensor type definition and Calibration.
- **Controls** – Configure logic hysteresis and setpoints.
- **Timers** – configure timers functions
- **WiConfig** – Manage Wi-Fi setup (Access Point or Station mode), Email contact for smart and AI functions
- **Sentinel** – Safety monitoring with min/max thresholds, persistence time and alarm detection
- **Reporter** – Automatic status and alarm reports via email, with configurable schedules
- **NetSky** – AI-assisted diagnostics, smart suggestions and system interaction
- **About** – Displays system information and useful links.

Monitor Page

(1)

(2)

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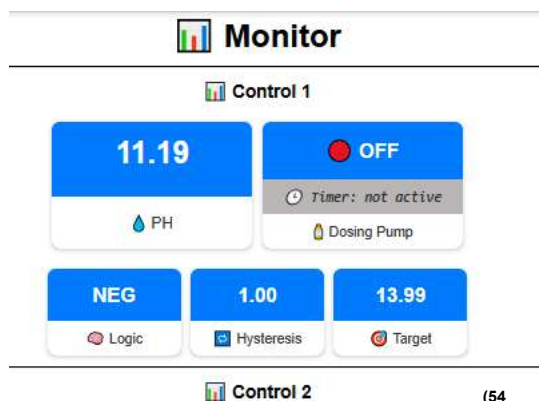


Fig. 6 Monitor Page

This page displays the current status of the device, including:

- (1) Current sensor reading
- (2) Current state of the actuator (ON/OFF)
- (2a) Actuator Status
- (2b) Timer status

Note: The actuator will be ON **only** if the control logic is active (e.g., pH below the threshold in positive logic) **and** the timer is either disabled or currently in its ON phase.

This allows **time-based modulation** of the actuator using the timer function, even when the control condition remains continuously true.

- (3) Regulation logic (Negative: actuator tries to reduce the current value till the set point; Positive: actuator tries to increase the current value till the set point)
- (4) Hysteresis (range of no action across the set point)
- (5) Set point setpoint

The same logic and conditions apply also to **Control 2, 3, 4**.

Sensors

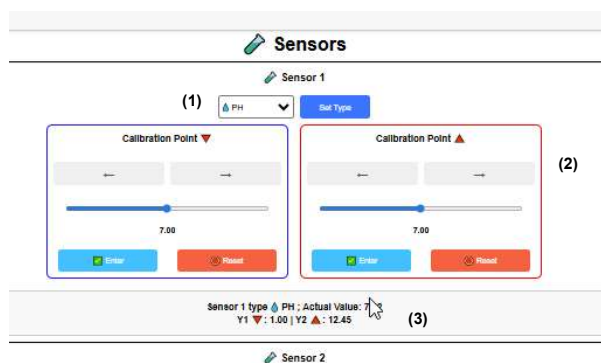


Fig. 7 Calibration Page

The Calibrator section is used to calibrate the connected sensor (pH Redox EC). It consists of two main elements:

- (1) Sensor Type Setting
Allows selecting the type of sensor to calibrate (pH or Redox).
- (2) Calibration Input Area
Used to enter the reference values of the calibration solutions for each calibration points (Low ▼ or High ▲).
- (3) Message Box
Displays:

the current sensor type and live measured value, and the currently stored calibration points (Low ▼ or High ▲).

How to Perform a Calibration

To carry out the calibration, prepare two standard buffer solutions. Then press Enter to begin. The message box will guide you through the following steps:

- I. Insert the sensor into the first buffer solution, then press Enter.
 - II. Enter the known value of the first solution and press Enter.
 - III. Wait approximately 10 seconds for the reading to stabilize, then press Enter.
 - IV. Insert the sensor into the second buffer solution and press Enter.
 - V. Enter the known value of the second solution and press Enter after waiting around 10 seconds for stabilization.
- Once all steps are completed, the calibration will be successfully saved and applied.

Note:

- To cancel the calibration process at any time, press the **Restart button**.
- To restore the factory calibration parameters, press the **Reset button**.

Note:

Using only a single calibration point may introduce measurement errors, as the system cannot fully compensate the sensor slope.

For this reason, it is strongly recommended to always perform a two-point calibration, selecting reference values as far apart as possible, in order to achieve the highest accuracy and stability.

The same calibration logic applies to **Sensors 3 and 4** when configured as measurable process sensor (**Ph Redox or EC**). **Sensor 2** is dedicated to temperature measurement only and does not require calibration.

If **Sensor 3** is configured for temperature-based operation (standard, differential, or average mode), calibration is not required.

Controls

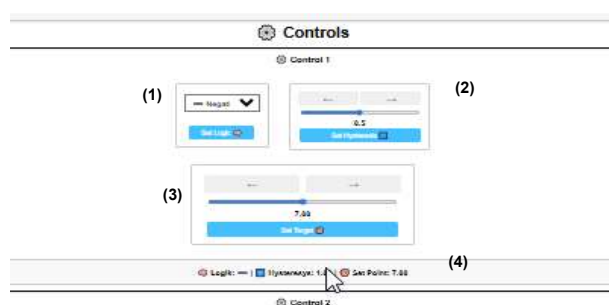


Fig. 8 Settings Page

This page allows the user to configure the control behavior through the following sections:

- (1) Logic Setting Controls
Define whether the actuator works in Negative (tries to reduce the value) or Positive logic (tries to increase the value).
 - (2) Hysteresis Setting Controls
Set the hysteresis range around the set point where no action is taken, avoiding rapid ON/OFF toggling.
 - (3) Target Setting Controls
Define the desired set point for the sensor (e.g., target pH or Redox value).
 - (4) The selected settings are immediately reflected in the status panel below, as shown in the figure.
- By pushing the buttons in each section, the configuration can

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be adjusted in real time.

The same logic and conditions apply also to **Control 2, 3, 4**.

Timers

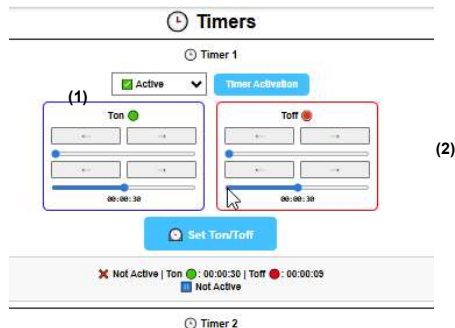


Fig.9 Timers Page

In the **Timers Page**, it is possible to configure the timer-based control of the actuator:

(1) Enable or disable

To activate or deactivate the timer, use the dropdown menu at the top, and press the **"Timer Activation"** button to apply the selected state.

(2) Set the **TON** time (the ON duration) using the scroll bars and adjustment buttons.

(3) Set the **TOFF** time (the OFF duration) in the same way. After adjusting the values, press the **"Set TON/TOFF"** button to apply the configuration.

(4) The **Status Panel** at the bottom of the page provides a clear summary:

-(4a) The first row shows the current timer settings: whether it is **Active** or **Not Active**, and the configured **TON** and **TOFF** times.

-(4b) The second row displays the **current state of the relay** (ON or OFF), based on the timer cycle and control condition.

→ The same configuration logic applies to **Timer 2**, which is associated with **Control 2, 3 4**.

WiConfig

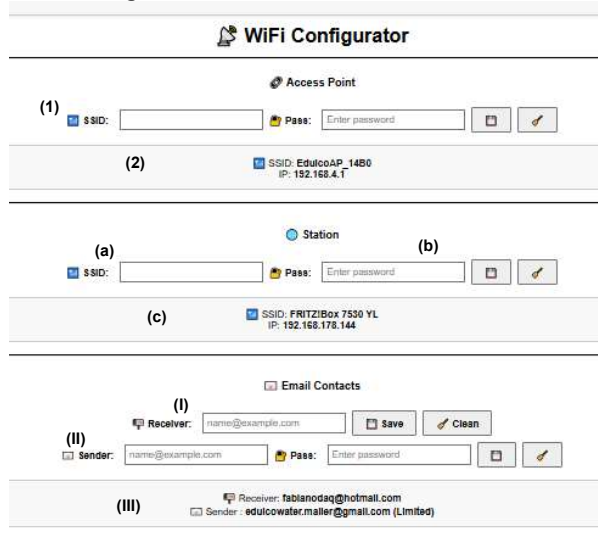


Fig.10 WiFi configuration Page

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or even through the internet worldwide.

(1) WiFi Credentials Input

In this section, you can enter the credentials (SSID and Password) of your private WiFi network.

By pressing the Save button, the device will attempt to connect to the router using the provided information.

(2) Access Point Status and Information

This section displays information related to the device's Access Point mode. When the device operates as an Access Point, it shows the SSID (network name) being broadcast, the IP address to be used for connection, and other relevant connection parameters. This allows direct configuration without requiring an external WiFi network.

(3) Station Mode Configuration and Status

This section allows you to save the credentials (SSID and Password) of your private WiFi network (router). Once saved, the device will attempt to connect to the specified network.

The panel displays the current connection status. When successfully connected, the assigned IP address will be shown.

At this point, the web interface can be accessed from any device connected to the same WiFi network by entering the shown IP address in a browser.

By pressing the clean button (🗑️) is possible to cancel the stored (🔑) credentials.

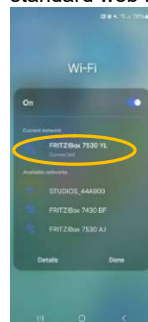
To connect the device to a local WiFi router:

(a)-Enter the **SSID** and **Password** of your private WiFi network in the designated fields.

(b)-Press the **Save** button to initiate the connection process.

Once connected, the system will automatically obtain an **IP address** from the router (c). This IP address will be displayed in the **message section**.

After the connection is established, any device (such as a smartphone, tablet, or PC) connected to the **same WiFi network** can access the **Studios Web Page Interface** using a standard web browser.



Simply enter the assigned IP address into the browser's address bar, as shown in the example.

Multiple devices can connect to and interact with the StudiosManager interface **simultaneously**.



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“Email Contacts”

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or even through the internet worldwide.

The WiFi Configurator allows the device to connect to a private WiFi network for access via a local router or worldwide through the internet.

(I) In this section you can define the **recipient email address** used by all advanced functions such as **Sentinel**, **NetSky**, and **Reporter**, which are described in the following sections.

(II) Here you can configure the **sender email account (SMTP/IMAP service)** used to send notifications. By default, the system is preconfigured with the **edulcowater.mailer@gmail.com**, which is limited to **5 emails**. For **privacy reasons** and **unlimited usage**, it is recommended to configure your **own email account** by following the guide below in the next “**Note**” chapter.

(III) The **status panel** reflects the current state of the **email configuration and settings**.

NOTE: How to create a private gmail sender email

Email Configuration (Gmail Setup)

To allow your EducoWater device to send alert messages via Gmail, you need to create a dedicated Gmail account and enable a special **App Password**.

Follow these steps carefully:

1 Create a Gmail account

It is recommended to use a dedicated account (for example: **edulcowater.alerts@gmail.com**) instead of your personal one.

2 Enable 2-Step Verification

-Go to <https://myaccount.google.com/security>

-In the section “**Signing in to Google**”, click **2-Step Verification**

-Follow the steps to activate it (you’ll need your phone).

3 Generate an App Password

After 2-Step Verification is enabled:

-Return to the same security page

-Click **App passwords** (or go directly to

<https://myaccount.google.com/apppasswords>)

-Sign in if requested

-Under **Select App**, choose **Mail**

-Under **Select Device**, choose **Other (Custom name)** and type:

“MyEducoWaterSender”

-Click **Generate**

4 Copy the generated password

A yellow window will appear with a 16-character password, for example:

“abcd efgh ijkl mnop”

Copy it **without spaces** → “abcdefghijklmnop”

5 Enter your credentials in the EducoWater web interface

-Open the configuration page

-In the field **Sender**, type your Gmail address (e.g.

edulcowater.alerts@gmail.com)

-In the field **Pass**, paste the 16-character App Password (without spaces)

-Press **Save**

6 Restart or reconnect the Email service

The device will store your credentials in memory and automatically reconnect to Gmail’s SMTP server.

Important Notes

-Regular Gmail passwords **do not work**. You must use the App Password.

-Do **not share** the App Password with others.

-If you change or delete it from your Google account, generate a new one and re-enter it in the device.

-Each App Password can be revoked anytime from your Google Account → **Security** → **App Passwords**.

Remote Access (Worldwide):

If **port forwarding** is configured on your router, the system can also be accessed **remotely from anywhere in the world**. Please consult your internet service provider or router documentation for instructions on how to enable and configure port forwarding.

As an example u can find one explaining link for the FritzBox router

<https://www.youtube.com/watch?v=uSC1-iWqibE>

In the following picture there is an illustration of all the possible ways to access the device.

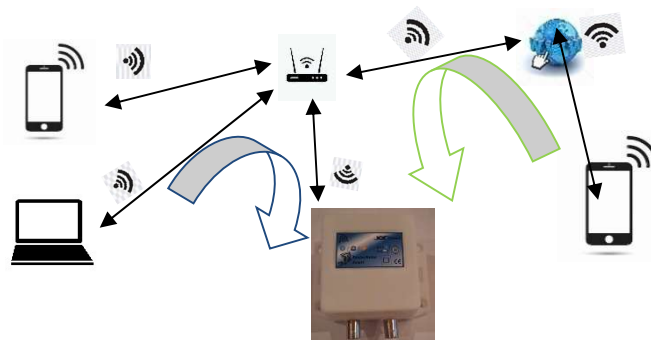


Fig.11 Remote mode WiFi connection

Sentinel

Fig.12 Sentinel Page

The Sentinel function continuously monitors a control value and sends alert emails to the address configured in the WiFi Configurator whenever the value goes out of the allowed limits. It is designed to detect abnormal conditions and notify you reliably, avoiding false alarms.

(1) Limits configuration

Using the up and down arrows, you can set the minimum and maximum allowed values.

If the measured value goes outside these limits, the condition is considered out of range.

(2) Timing configuration

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Warning Time: the time that the value must remain out of limits before it is considered a real alarm (filters short spikes).

Notification Interval: how often the alarm email is repeated if the alarm condition persists.

(3) Activation and save

With the activation button, the Sentinel can be set ACTIVE or NOT ACTIVE.

When active, the Sentinel starts monitoring using the configured limits and timings.

(4) Status panel

The status panel reflects the current Sentinel configuration and state, showing whether it is active and how it is operating.

If the monitored value **returns within the configured limits**, the Sentinel automatically sends a **recovery email** to inform that the alarm condition has been resolved.

This guarantees a **complete notification cycle**: alarm raised → alarm cleared.

Below one example of Alarm notification and alarm resolved notification

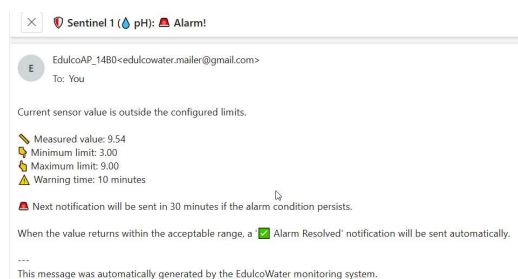


Fig. 13 Sentinel Alarm notification email

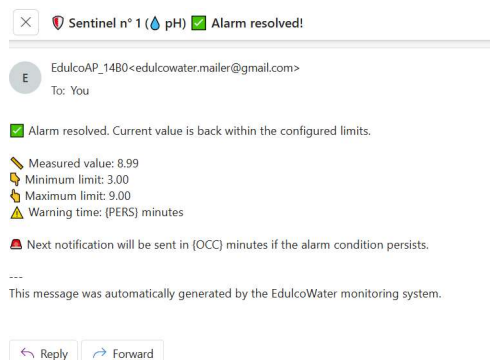


Fig. 14 Sentinel Alarm resolved notification email

The same applies to Sentinels 2, 3, 4.

Reporter

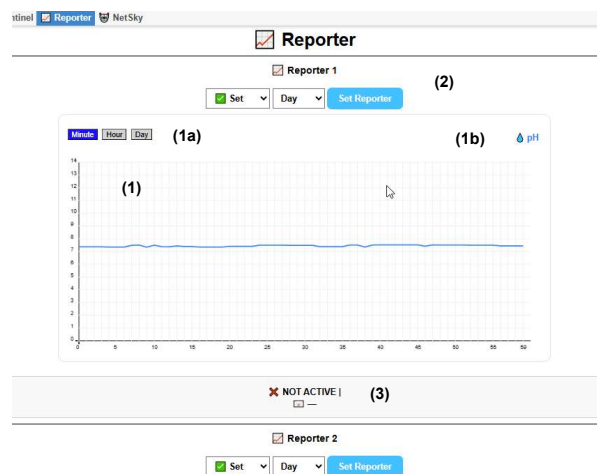


Fig. 15 Reporter Page

In this page you can monitor, in real time, the trend of the selected control value (pH, Redox, Temperature, etc.) and review its history over the last minute, hour, or day.

You can also configure the Reporter to send periodic reports by email, according to your selected settings (recipient email is defined in the WiFi Configurator).

(1) Trend Graph

Here you can view the recorded values of the control.

(1a) Display range selector: choose the visualization interval (minute / hour / day).

(1b) Legend: indicates the control type and what is being plotted.

(2) Report Settings

Select the desired report type and configure whether you want the report to be sent according to your chosen schedule/settings.

(3) Status Panel

The status panel reflects the current reporting state, including the data collection progress (%) before the next report is sent to the email configured in the WiFi Configurator.

In the following pictures an example of the reporting email and of the reported data in excell file format

Fig. 16 Reporter Email notification

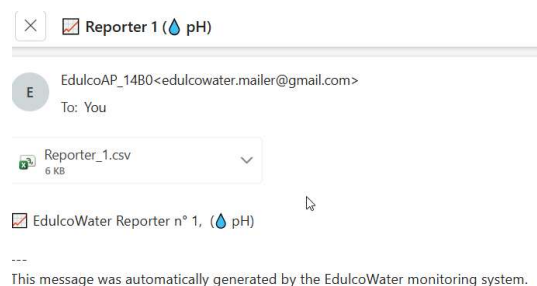


Fig. 17 Excell data reported example

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# TYPE: PH	A	B	C	D	E
Timestamp	Average	ValMin	ValMax		
12/13/2025 15:21	777	777	777		
12/14/2025 15:21	777	777	777		
12/15/2025 15:21	777	777	777		
12/16/2025 15:21	774	774	774		
12/17/2025 15:21	774	774	774		
12/18/2025 15:21	777	777	777		
12/19/2025 15:21	775	774	777		
12/20/2025 15:21	774	774	774		
12/21/2025 15:21	775	774	776		
12/22/2025 15:21	776	776	776		
12/23/2025 15:21	776	776	777		
12/24/2025 15:21	777	777	777		
12/25/2025 15:21	777	777	777		
12/26/2025 15:21	777	777	777		
12/27/2025 15:21	777	777	777		
12/28/2025 15:21	777	777	777		
12/29/2025 15:21	777	777	777		
12/30/2025 15:21	777	777	777		
12/31/2025 15:21	777	777	777		
1/1/2026 15:21	777	777	777		

NetSky


Send Message (1a)

Hello EducoWater copilot system,
I have no time to go in the instruction manual and check.... boring...
Can u tell me how can i set the ph of controller 1 to 7.56?
Thank u

(1)


Execute Commands (2a)

```
cmd=setcontrolsetpoint&ctrl=0&setpoint=7.56
```

(2)


Message sent (3)


SelfPoint set

Fig.18 NetSky Page

NetSky is the first system on the market to integrate an AI Copilot assistant directly into the device. By writing a message as shown in the example, you can receive intelligent assistance for diagnostics, user manuals, and other device-related information.

(1) Email Message

In this field you can write your request or description of the issue.

The message will be analyzed by the Copilot system to provide targeted assistance.

(1a) Send Button

Press the button to send the message by email and receive support from the NetSky Copilot system.

(2) Prompt commands

In this field you can copy and paste commands the NetSKY system suggests to u.

(2a) Commands Execution Button

Press the button to set the copied and pasted commands in order to adjust the EducoWater according to your needs.

Fig.19 NetSky Email response


Re: EducoWaterCopilotRequest

--- NetSky EducoWater AI assistant: ---

To set the pH of controller 1 to 7.56, you can do it in these ways:

Using the display:

- Move the cursor to the setpoint parameter.
- Press MODE so the cursor blinks.
- Use UP/DOWN to set the value to 7.56.
- Press MODE again to save.

(1)

Using the web server:

- Go to the Settings page for Control 1.
- Set the target (set point) to 7.56.
- Apply/save the setting.

Using smart home command (Bash/HTTP POST):

Send this command:

```
curl -X POST http://192.168.178.109/smartcommand -d "cmd=setcontrolsetpoint&ctrl=0&setpoint=7.56"
```

(Here, 7.56 means 7.56)

Answer confidence level: SURE

(3)


Answer to message (4)

From: EducoAP_1480

Original Message:

Heito EducoWater copilot system, I have no time to go in the instruction manual and check.... boring... Can u tell me how can i set the ph of controller 1 to 7.56? Thank u

In the reply email, the following sections are included:

(1) Explanation

Explanatory text with direct reference to the commands shown on the display and their meaning.

(2) Bash / Control Commands

Examples of executable commands (e.g. for Smart Home integration) that can be copied and used directly.

(3) Confidence Level

An indication of the confidence level of the response, meaning how reliable the system considers the proposed solution (yeah... it's still growing 🌱).

(4) Request Details

Information related to the original request message, for traceability and context.

About


EducoWater-Premium

EducoWater-Premium is a Wi-Fi-enabled device for water quality monitoring and automation. It features two independent automatic pumps, dual Wi-Fi channels (Access Point and Station modes), and an integrated LCD display for intuitive local control. The system supports pH and ORP sensors, remote management via web interface, automated dosing logic, and modular relay control. Ideal for swimming pools, hydroponics, systems, aquariums, and similar environments where precise and reliable water management is essential.

Device Information

Device Type: EducoWater-Premium

WiFi: 01:00

SW: 01:00

Device ID: ESTAD4

You can purchase EducoWater devices at:

- "Smart" Line on eBay.de
- "Smart" Line on eBay.it
- "Modular" Line on eBay.de
- "Modular" Line on eBay.it
- "Modular Cabinet" Line on eBay.de
- "Modular Cabinet" Line on eBay.it
- "Industrial" Line on eBay.de
- "Industrial" Line on eBay.it
- "Premium" Line on eBay.de
- "Premium" Line on eBay.it

Visit the StudioS official web site

www.studiosds.com

EducoWater is also on

Facebook.com/EducoWater

For questions contact us here (don't change the subject otherwise ur request will be ignored)

studios_ds@outlook.de

Fig.20 About Page

The About section provides a brief description of the product and useful resources for further support or information.

Product Overview Useful Links like e commerce company web site and email for technical support

Technical Data:

Electrical Ratings

Descritption	Rating
Supply	5Vdc
Power consumption	1W
WiFi Range (1)(2)	10mt

Sensorsand Controls Ratings

Descritption	Rating
Ph range	0.01-14.00

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 Ph error	±0,01
 Ph resolution	±0,01
 Ph Control algorithm	ON-OFF with Adjustable Hyster and Logic
 Redox range	-1800/1800 mV
 Redox error	±1 mV
 Redox resolution	±1 mV
 Redox Control algorithm	ON-OFF with Adjustable Hyster and Logic
 Temp Range	5°C-70°C
 Temp error	±1 °C
 Temp resolution	±0.1°C
 Temp Control algorithm	ON-OFF with Adjustable Hyster and Logic
 EC Range	10 – 20,000 µS/cm
 EC error	±10 µS/cm
 EC resolution	±1 µS/cm (10–2,000 µS/cm), ±10 µS/cm (2,000–20,000 µS/cm)
 EC Control algorithm	ON-OFF with Adjustable Hyster and Logic
 TDS Range	6 – 12,800 ppm
 TDS error	±7 ppm
 TDS resolution	1 ppm (6–1,280 ppm), 10 ppm (1,280–12,800 ppm)
 TDS Control algorithm	ON-OFF with Adjustable Hyster and Logic
 Salinity Range	6 – 11,000 ppm
 Salinity error	±6 ppm
 Salinity resolution	1 ppm (6–1,100 ppm), 10 ppm (1,100–11,000 ppm)
 Salinity Control algorithm	ON-OFF with Adjustable Hyster and Logic
 Timer Range	1 second – 24 hours
 Timer Resolution	1 second
 Timer Accuracy	±10 sec/day
 Timer Function	Cyclic ON/OFF output modulation

(1) The range can be extended in **Access Point mode** by using any standard WiFi repeater, and in **Station mode** through any router, whether connected to the internet or not.

Mechanical Overall Dimensions:
90x65x40

Markings: CE

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SMART HOME FUNCTIONS

Using Smart Home Commands

The Smart Home commands of the EducoWater Premium system are designed to enable seamless integration with external platforms such as Amazon Alexa, Node-RED, Home Assistant, or any custom automation system that supports HTTP POST requests. These commands allow full remote access to the control logic, timers, calibration functions, and system configuration.

In the example shown below, a command is executed using a bash terminal to interact with the device through its local IP address.

See the figure below for a real example of sending a command from Bash (Linux/macOS/Windows terminal with curl).

To run a command from the terminal:

Open a terminal or command prompt on your PC or smart device.

Type the provided curl command as shown in the examples.

Replace the IP address with your device's actual IP (in the example it was **192.168.178.130** but u can get your own in the WiFiConfiguration page panel statu).

Press ENTER to send the command.

MINGW64:/c/Fabiano/05_Projects/03_Studios/53_Edulcowater-Smart/70_Manuals

```
DAQUFAB@K123716H1 MINGW64 /c/Fabiano/05_Projects/03_Studios/53_Edulcowater-Smart/70_Manuals
$ curl -X POST http://192.168.178.130/smartcommand -d "cmd=getdeviceabout"
% Total % Received % Xferd Average Speed Time Time Time
100 177 100 159 100 18 1715 194 --:-- --:-- --:-- 1923{"deviceType":"EducoWater-Premium","hwVersion":"01.00","swVersion":"01.00","deviceId":"E07A04","ssid":"Educo_asdf","password":"123456789","ip":"192.168.4.1"}
DAQUFAB@K123716H1 MINGW64 /c/Fabiano/05_Projects/03_Studios/53_Edulcowater-Smart/70_Manuals
$ |
```

Implemented Commands

The following pages describe in detail all the GET and SET commands currently implemented in the system.

Each command includes a description, required parameters, limitations, example responses, and a working Bash curl example.

GET Commands

-  **getdeviceabout** — Retrieve general device information
-  **getcontrolstatus** — Get the configuration and live status of a control channel
-  **getcalstatus** — Retrieve calibration data and the current sensor value
-  **getsystemstatus** — Get the operating mode, manual status, and system timer configuration

SET Commands

-  **setsystemmode** — Set the operating mode to Manual, Timer, Automatic, or SmartHome
-  **setcontroltype** — Set a supported control type for a compatible control channel
-  **setcontrolhyster** — Define the control hysteresis value
-  **setcontrollogik** — Configure direct or inverse control logic
-  **setcontrolsetpoint** — Define the control setpoint
-  **settimeractivation** — Enable or disable a timer block
-  **settimerontoff** — Set the timer ON and OFF durations
-  **resetcalibration** — Reset a supported sensor calibration point
-  **setsmarthomestatus** — Turn the Smart Home manual output ON or OFF
-  **setsystemtimer** — Set the global system timer ON and OFF durations

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1) GET DEVICE ABOUT

Description

This command retrieves general information about the EducoWater device. It returns the device identity, firmware and hardware versions, and Access Point network information.

Required Parameters

none

Limitations

a) None

Response

```
{ "deviceType": "EducoWater-Plattform", "hwVersion": "01.00", "swVersion": "01.00", "deviceId": "A1B2C3", "ssid": "EDULCO_1",
  "password": "mypassword", "ip": "192.168.4.1" }
```

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getdeviceabout"
```

2) GET CONTROL STATUS

Description

This command retrieves the current status and configuration of a specific control channel. It returns control metadata (type and linked sensor), current measured value, control settings (setpoint, hysteresis, logic), actuator status, and the associated timer status (if used).

Required Parameters

ctrl: Control index (must be 0)

Limitations

a) Only control 0 is supported

Response

```
{ "ctrl": 0, "type": "ph", "active": true, "logic": false, "setpoint": 703, "hysteresis": 25 }
```

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getcontrolstatus&ctrl=0"
```

3) GET CALIBRATION STATUS

Description

This command retrieves the calibration data and current measurement value of a specified sensor. It returns the raw calibration points stored in EEPROM, along with the current sensor reading and sensor type.

Required Parameters

sensor: Sensor index 0

Limitations

a) Only sensor index 0

Response

```
{ "sensor": 0, "type": 3, "value": 977, "calX1": 709, "calY1": 100, "calX2": -644, "calY2": 1245 }
```

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getcalstatus&sensor=0"
```

4) GET SYSTEM STATUS

Description

This command retrieves the current global operating status of the device. It returns information about the working mode, manual control status, and system timer configuration. This command is useful for remote monitoring and smart home integration.

Required Parameters

None.

Limitations

None.

Response

```
{ "mode": "Automatic", "manual": false, "timer": { "active": true, "timeOn": 300, "timeOff": 120 } }
```

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=getsystemstatus"
```

Exception

This command is available only on models that support Manual mode.

5) SET SYSTEM MODE (Working Mode)

Description

This command sets the global working mode of the device. The system mode defines how the device controls the output:

- Manual // Timer Mode // Automatic Mode // SmartHome

Required Parameters

mode : Desired system mode (string, case-sensitive); "Manual", "Timer", "Automatic", "SmartHome"

Limitations

- a) mode must be exactly one of: "Manual", "Timer", "Automatic" "SmartHome"
- b) If mode is missing → the command returns an error.

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✓ Response

✓ System mode updated to <mode> // ✗ Failed to set system mode

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setsystemmode&mode=Manual"
```

⚠ Exception

This command is available only on models that support Manual mode.

6) 🚀 SET CONTROL TYPE

📄 Description

This command sets the sensor type associated with a specific control index. Supported types are **pH ORP and Temperature**.

📋 Required Parameters

ctrl: Control index 0 // type: Desired type "ph" "orp" "ntc" "ec" "tds" "sal" "diff" "avg"

⚠ Limitations

- a) Only control index 0 or 3 supported
- b) Type must be exactly "ph" "orp" "ntc" "ec" "tds" "sal" "diff" "avg"

✓ Response

✓ Control type set // ✗ Failed to set control type

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolype&ctrl=0&type=ntc"
```

⚠ Exception

Available control channels and supported control types depend on the device model. A control type can be assigned only to a channel compatible with the corresponding sensor or input. Refer to the model-specific control compatibility table for the supported channel/type combinations.

7) 🔄 SET CONTROL HYSTERESIS

📄 Description

This command sets the hysteresis value for a specific control channel. The hysteresis defines the tolerance band around the setpoint to prevent fast toggling of the output.

📋 Required Parameters

ctrl: Control index 0 //hysteresis: Value in ADC units (integer)

⚠ Limitations

- a) Only control index 0 supported
- b) Hysteresis must be a positive integer

✓ Response

✓ Hysteresis set // ✗ Failed to set hysteresis

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolhyster&ctrl=0&hysteresis=49"
```

8) 🧠 SET CONTROL LOGIC

📄 Description

This command sets the logic mode of a control channel. The logic determines how the relay output reacts relative to the setpoint: **true** → Direct logic; **false** → Inverse logic The exact switching behavior depends on the configured control type (pH, ORP, etc.).

📋 Required Parameters

ctrl: Control index 0 //logic: "true" or "false"

⚠ Limitations

- a) Only control index 0 supported
- b) Logic must be "true" or "false"

✓ Response

✓ Logic set // ✗ Failed to set logic

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrollogik&ctrl=0&logic=false"
```

9) 🎯 SET CONTROL SETPOINT

📄 Description

This command sets the target setpoint value for a specific control channel.

📋 Required Parameters

ctrl: Control index (must be 0) // setpoint: Desired setpoint value (integer)

⚠ Limitations

- a) Only control index 0 supported
- b) Setpoint must be a valid integer in ADC range

✓ Response

✓ SetPoint set // ✗ Failed to set SetPoint

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setcontrolsetpoint&ctrl=0&setpoint=704"
```

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10) SET TIMER ACTIVATION

Description

This command activates or deactivates a timer block. Useful for enabling/disabling periodic logic based on time cycles.

Required Parameters

timer: Timer index 0 // active: "true" or "false"

Limitations

- a) Timer index must be 0
- b) Active must be "true" or "false"

Response

✓ Timer activation set // ✗ Failed to set timer activation

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=settimeractivation&timer=0&active=false"
```

11) SET TIMER TON/TOFF

Description

This command sets the ON (TON) and OFF (TOFF) durations in seconds for a selected timer. The timer runs cyclically: ON for ton seconds; OFF for toff seconds

Required Parameters

timer: Timer index 0 // ton: Duration in seconds for ON state // toff: Duration in seconds for OFF state

Limitations

- a) ton and toff must be between 1 and 64859 seconds
- b) Command accepted only if device is in TIMER mode

Response

✓ TON and TOFF set // ✗ Failed to set TON or TOFF

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=settontoff&timer=0&ton=180&toff=240"
```

12) RESET CALIBRATION

Description

This command resets a specific calibration point of a sensor to its default value. It is typically used to restore factory calibration or to restart the calibration procedure. Each sensor uses two calibration points, identified by the parameter calpoint.

Required Parameters

sensor: Sensor index 0 // calpoint: 0 → First calibration point; 1 → Second calibration point

Limitations

- a) Only valid sensor indexes are accepted by the firmware.

Response

✓ Calibration reset to default // ✗ Reset failed

Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=resetcalibration&sensor=0&calpoint=1"
```

Exception

This command does not apply to control types that do not support two-point calibration, such as ntc, temp, diff, and avg. It is available only for sensors and control types that support calibration points.

13) SET SMART HOME MODE

Description

This command sets the device in Smart Home mode. It is used to turn the manual control output ON or OFF from a smart home master like "home Assistant"

Required Parameters

status : Desired Relay state (string, case-sensitive); "on" → Relay output ON, "off" → Relay output OFF.

Limitations

- a) status must be exactly "on" or "off".
- b) If status is missing → the command returns an error.

Response

✓ Manual status updated // { "error": "Missing status parameter" }

Example (bash curl)

```
curl -X POST http://192.168.178.170/smartcommand -d "cmd=setsmarthomestatus&status=on"
```

Exception

This command is available only for controls that include and support the Manual control page.

14) SET SYSTEM TIMER

Description

This command sets the global system timer ON and OFF durations. The system timer is used when the device operates in Timer mode, causing the output to switch cyclically between ON and OFF states.

Required Parameters

timeon : ON duration in seconds (integer) // timeoff : OFF duration in seconds (integer)

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⚠ Limitations

- a) Both parameters are required; missing values will return an error.
- b) Valid time ranges depend on the internal firmware logic.

✅ Response

- ✅ Timer updated // { "error": "Missing timeon/timeoff parameters" }

💡 Example (bash curl)

```
curl -X POST http://192.168.178.130/smartcommand -d "cmd=setsystemtimer&timeon=300&timeoff=120"
```

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PLC FUNCTIONS: MODBUS TCP REFERENCE

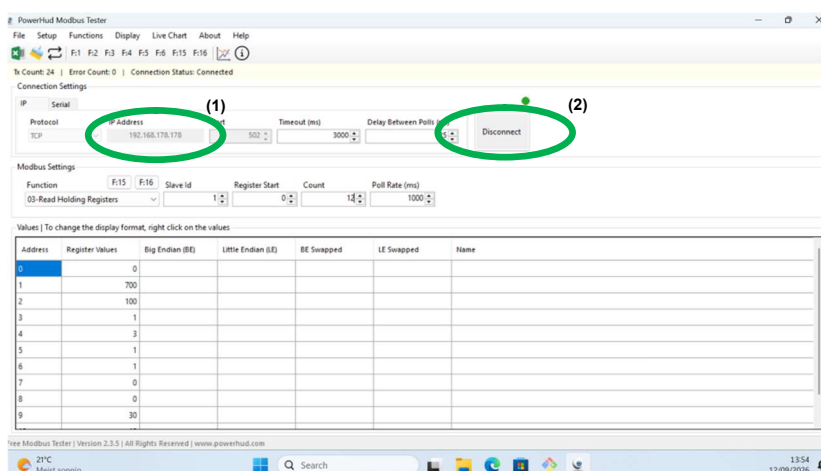
Using PLC commands

The PLC functions of the EducoWater system are designed to enable seamless integration with industrial automation systems, **PLCs**, **HMIs**, **SCADA systems**, and other devices supporting the **Modbus TCP protocol**.

EducoWater operates as a **Modbus TCP Server (Slave)**, while the connected PLC or automation controller operates as the **Modbus TCP Client (Master)**. This allows external automation systems, including **Schneider Electric Modicon M221/M241/M262** and **Siemens S7-1200/S7-1500 PLCs**, to read process values and device status and to write control parameters directly through the Ethernet/Wi-Fi network.

The Modbus TCP interface uses **TCP port 502**, **Unit ID / Slave ID 1**, and Holding Registers. **FC03 (Read Holding Registers)** is used for reading; **FC06 (Write Single Register)** for writing one register; and **FC16 / 0x10 (Write Multiple Registers)** for writing multiple consecutive registers.

For development, commissioning, and diagnostic purposes, it is not necessary to have a physical PLC available. The Modbus TCP interface can be tested using any standard Modbus TCP Master/Client testing software, such as the Modbus Tester **ModbusTester 3.2.0** shown in the figure below and available at this link <https://www.modbustester.com/>



Once the Modbus testing software has been installed, launch the application and configure the connection using the IP address of the **EducoWater device (1)**

The EducoWater device and the **PLC**, or the PC running the PLC/Modbus Master simulator, must be connected to the **same local network/router**.

Enter the EducoWater IP address, select **Modbus TCP**, set the communication port to **502** and the **Unit ID to 1**, then press **Connect (2)** to establish the connection with the device.

Once connected, the PLC or Modbus testing software can read and write the parameters exposed by EducoWater through the Modbus TCP interface.

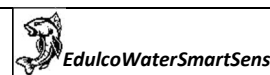
MODBUS TCP REGISTER MAP

The EducoWater Industrial Modbus TCP interface provides **60 Holding Registers**, addressed from 0 to 59. The first **48** registers are divided into **4 control** blocks of **12** consecutive registers. Registers **48–49** are **reserved**, and registers **50–59** contain **system parameters**.

Addressing Note: The protocol addresses below are zero-based. In 4xxxx notation, address 0 corresponds to 40001, address 1 to 40002, and so on.

The following **Control Registers Table** provides a complete overview of all Modbus control registers available in EducoWater, including register addresses, access permissions, data types, units, descriptions, and compatibility with the various EducoWater device models

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Control Registers Table

Register	Address	Control	Parameter	Description	Access	Device Compatibility
0	40001	0	ActReadVal	Current value read from the sensor associated with the control.	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
1	40002	0	Set_Point	Control setpoint used by the regulation logic.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
2	40003	0	Set_Hyster	Hysteresis value applied around the configured setpoint.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
3	40004	0	CtrlSt	Current control/relay output state. 0 = OFF, 1 = ON.	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
4	40005	0	type	Type of control/sensor associated with the control channel.	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
5	40006	0	Active	Enables/disables the control. 0 = Disabled, 1 = Enabled.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
6	40007	0	Set_PosLogik	Control logic configuration / operating direction.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
7	40008	0	HwRelay	Hardware relay associated with the control. Index 0–3.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
8	40009	0	Timer.Active	Enables/disables the control timer. 0 = Disabled, 1 = Enabled.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
9	40010	0	Timer.OnSeconds	Timer ON duration in seconds.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
10	40011	0	Timer.OffSeconds	Timer OFF duration in seconds.	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
11	40012	0	Timer.ElapsedTime	Current elapsed timer time.	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
12	40013	1	ActReadVal	Same as for control 0 register 0	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
13	40014	1	Set_Point	Same as for control 0 register 1	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
14	40015	1	Set_Hyster	Same as for control 0 register 2	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
15	40016	1	CtrlSt	Same as for control 0 register 3	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
16	40017	1	type	Same as for control 0 register 4	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
17	40018	1	Active	Same as for control 0 register 5	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
18	40019	1	Set_PosLogik	Same as for control 0 register 6	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
19	40020	1	HwRelay	Same as for control 0 register 7	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
20	40021	1	Timer.Active	Same as for control 0 register 8	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
21	40022	1	Timer.OnSeconds	Same as for control 0 register 9	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
22	40023	1	Timer.OffSeconds	Same as for control 0 register 10	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
23	40024	1	Timer.ElapsedTime	Same as for control 0 register 11	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-SENS-EC-TEMP
24	40025	2	ActReadVal	Same as for control 0 register 0	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
25	40026	2	Set_Point	Same as for control 0 register 1	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
26	40027	2	Set_Hyster	Same as for control 0 register 2	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
27	40028	2	CtrlSt	Same as for control 0 register 3	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
28	40029	2	type	Same as for control 0 register 4	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
29	40030	2	Active	Same as for control 0 register 5	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
30	40031	2	Set_PosLogik	Same as for control 0 register 6	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP

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31	40032	2	HwRelay	Same as for control 0 register 7	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
32	40033	2	Timer.Active	Same as for control 0 register 8	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
33	40034	2	Timer.OnSeconds	Same as for control 0 register 9	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
34	40035	2	Timer.OffSeconds	Same as for control 0 register 10	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
35	40036	2	Timer.ElapsedTime	Same as for control 0 register 11	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-EC-TEMP; SMART-SENS-EC-TEMP
36	40037	3	ActReadVal	Same as for control 0 register 0	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
37	40038	3	Set_Point	Same as for control 0 register 1	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
38	40039	3	Set_Hyster	Same as for control 0 register 2	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
39	40040	3	CtrlSt	Same as for control 0 register 3	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
40	40041	3	type	Same as for control 0 register 4	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
41	40042	3	Active	Same as for control 0 register 5	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
42	40043	3	Set_PosLogik	Same as for control 0 register 6	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
43	40044	3	HwRelay	Same as for control 0 register 7	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
44	40045	3	Timer.Active	Same as for control 0 register 8	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
45	40046	3	Timer.OnSeconds	Same as for control 0 register 9	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
46	40047	3	Timer.OffSeconds	Same as for control 0 register 10	R/W	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
47	40048	3	Timer.ElapsedTime	Same as for control 0 register 11	R	INDUSTRIAL; MODULAR; PREMIUM; SMART-PLUG-PH-RX; SMART-SENS-PH-RX
48	40049	--	--	Reserved for future	--	ALL
49	40050	--	--	Reserved for future	--	ALL
50	40051	--	REG_SYSTEM_MODE	System mode: 0 = Manual, 1 = Timer, 2 = Automatic, 3 = SmartHome.	R/W	ALL
51	40052	--	REG_MANUAL_STATUS	General manual command status. 0 = OFF, 1 = ON.	R/W	ALL
52	40053	--	REG_SYSTEM_TIMER_STATUS	Current system timer status.	R	ALL
53	40054	--	REG_SYSTEM_TIMER_ON_SECONDS	System timer ON duration in seconds.	R/W	ALL
54	40055	--	REG_SYSTEM_TIMER_OFF_SECONDS	System timer OFF duration in seconds.	R/W	ALL
55	40056	--	REG_SYSTEM_TIMER_ELAPSED	Current elapsed time of the system timer.	R	ALL
56	40057	--	REG_SMARTHOME_STATUS	Smart Home status. 0 = OFF, 1 = ON.	R/W	ALL
57	40058	--	REG_SMARTHOME_CONNECTED	Smart Home connection status.	R	ALL
58	40059	--	REG_HW_TYPE	EducoWater application hardware type.	R	ALL
59	40060	--	REG_UNIT_ID	Modbus Unit ID. Current value: 1.	R	ALL

Implemented Commands

A brief explanation and practical examples are also provided for the supported Modbus functions:

- **FC06 – Write Single Register**
- **FC16 – Write Multiple Registers**

These examples can be used both for testing with a Modbus Master simulator and as a reference when configuring the communication with the actual PLC.

GET Commands

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1.  **FC03 – Read Holding Registers — Read one or more holding registers**
2.  **FC06 – Write Single Register — Write a value to a single holding register.**
3.  **FC16 – Write Multiple Registers — Write values to multiple holding registers in a single transaction.**

1) **FC03 Read Holding Registers**

- **Description**

The **FC03 – Read Holding Registers** function is used by the PLC or Modbus TCP Master to read one or more Holding Registers from the EducoWater device as per registers description in table “**Control Registers Table**”

- **Required Parameters**

EducoWater IP Address: Device local IP address;

TCP Port: **502**

Unit ID: **1**

Function Code: **03** – Read Holding Registers

Start Address: First register to be read

Quantity: Number of consecutive registers to read

- **Limitations**

- a) The requested Start Address must correspond to a valid EducoWater Holding Register.
- b) The requested address range must remain within the available Modbus Register Map.
- c) Registers 48 and 49 are reserved for future implementation.
- d) FC03 is a **read-only operation** and cannot be used to modify register values.

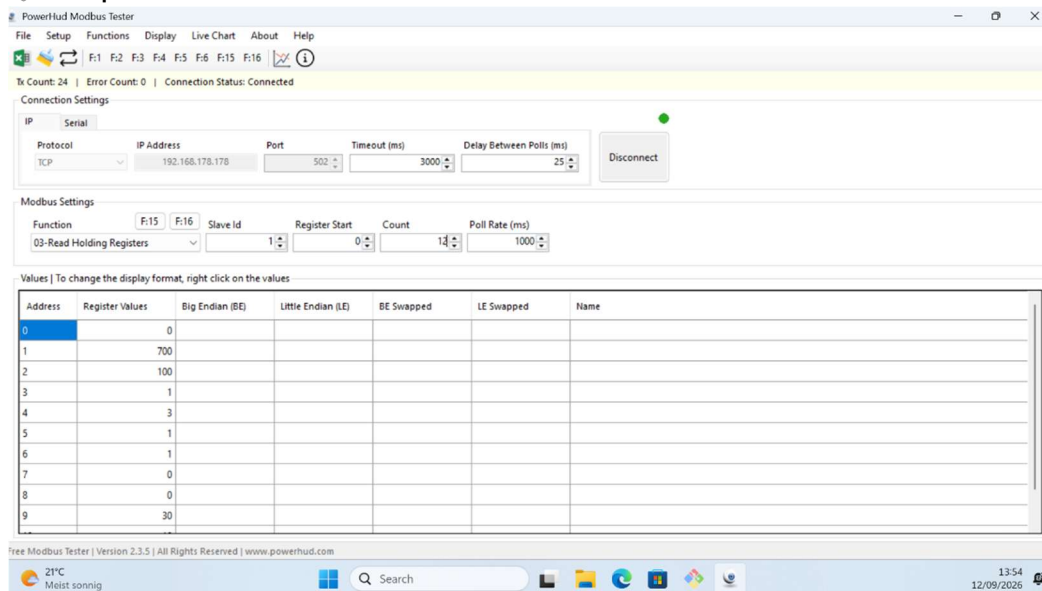
- **Response**

When the request is successfully executed, EducoWater returns the current value of each requested Holding Register.

For example, reading **Control 0** starting from Address 0 with a Quantity of 12 returns the complete Control 0 register block:

ActReadVal, Set_Point, Set_Hyster, CtrlSt, type, Active, Set_PosLogik, HwRelay, Timer.Active, Timer.OnSeconds, Timer.OffSeconds, and Timer.ElapsedTime.

Example – Modbus TCP Master



The screenshot shows the PowerHud Modbus Tester interface. The Connection Settings section is configured with IP Address 192.168.178.178, Port 502, Timeout 3000 ms, and Delay Between Polls 25 ms. The Modbus Settings section shows Function 03-Read Holding Registers, Slave ID 1, Register Start 0, Count 12, and Poll Rate 1000 ms. The Values table displays the following data:

Address	Register Values	Big Endian (BE)	Little Endian (LE)	BE Swapped	LE Swapped	Name
0	0					
1	700					
2	100					
3	1					
4	3					
5	1					
6	1					
7	0					
8	0					
9	30					

2) **FC06 Write Single Register**

- **Description**

The **FC06 – Write Single Register** function is used by the PLC or Modbus TCP Master to write a value to a single Holding Register of the EducoWater device as per registers description in table “**Control Registers Table**”.

Only registers identified as **Read/Write (R/W)** can be modified using this function.

- **Required Parameters**

EducoWater IP Address: Device local IP address;

TCP Port: **502**

Unit ID: **1**

Function Code: **06** – Write Single Register

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Register Address: Address of the Holding Register to be written

Value: New value to be assigned to the selected register

- ⚠ Limitations

- The requested Register Address must correspond to a valid EducoWater Holding Register.
- The selected register must be identified as Read/Write (R/W) in the Modbus Register Map.
- Registers identified as Read Only (R) cannot be modified using FC06.
- The written value must comply with the valid range and format defined for the selected register.

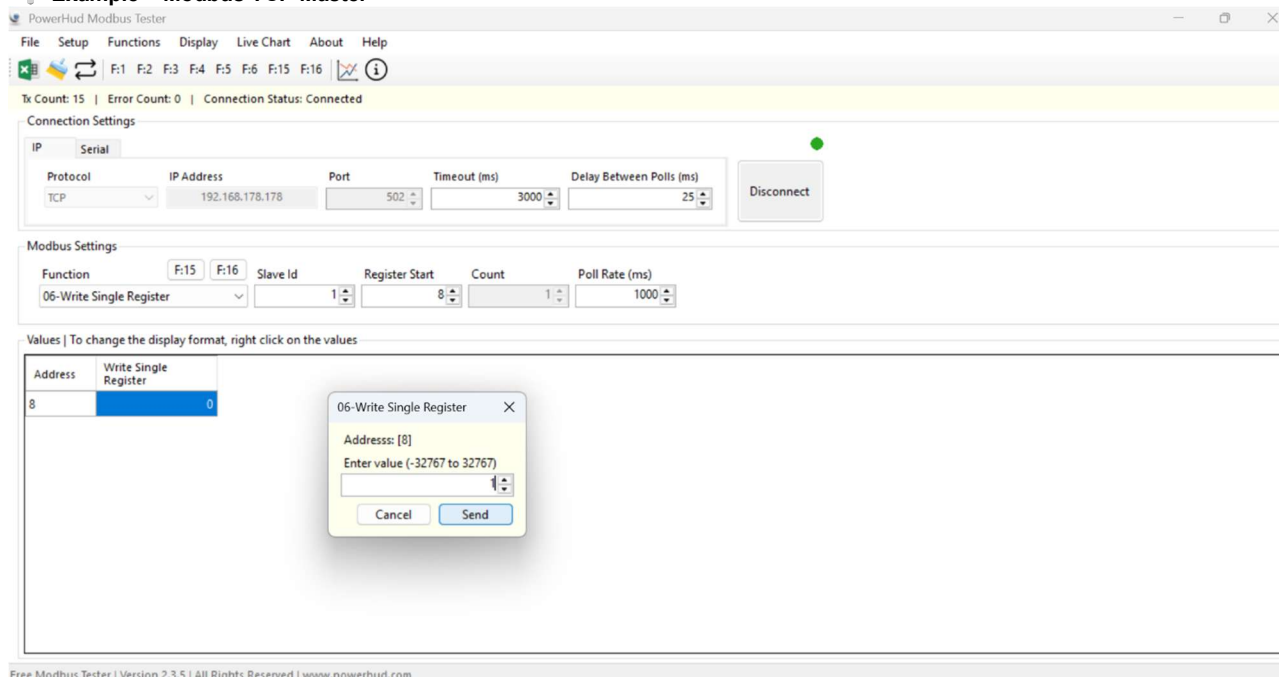
- ✅ Response

When the request is successfully executed, EducoWater writes the requested value to the selected Holding Register and returns a confirmation of the operation.

For example, writing to Address 1 modifies the Set_Point parameter of Control 0.

The updated value can subsequently be verified by executing an FC03 – Read Holding Registers request on the same register.

- 💡 Example – Modbus TCP Master



3) FC16 Write Multiple Registers

- 📖 Description

The **FC16 – Write Multiple Registers** function is used by the PLC or Modbus TCP Master to write multiple consecutive Holding Registers of the EducoWater device within a single Modbus TCP request, as per registers description in table “**Control Registers Table**”. Only registers identified as Read/Write (R/W) can be modified.

- 📋 Required Parameters

EducoWater IP Address: **Device local IP address**;

TCP Port: **502**

Unit ID: **1**

Function Code: **16 (0x10) – Write Multiple Registers**

Start Address: Address of the first Holding Register to be written

Quantity: Number of consecutive registers to be written

Values: Values to be assigned to the selected consecutive registers

- ⚠ Limitations

- The requested Start Address must correspond to a valid EducoWater Holding Register.
- The complete requested address range must remain within the available Modbus Register Map.
- Registers identified as Read Only (R) cannot be modified.
- Each written value must comply with the valid range and format defined for its corresponding register.
- The selected register range must contain only registers that are valid for the requested write operation.

- ✅ Response

When the request is successfully executed, EducoWater writes the requested values to the selected consecutive Holding Registers and returns a confirmation containing the starting address and the number of registers written.

For example, FC16 can be used to update multiple consecutive parameters of a Control block within a single Modbus transaction.

The updated values can subsequently be verified by executing an FC03 – Read Holding Registers request over the same address range.

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- Example – Modbus TCP Master

PowerHud Modbus Tester

File Setup Functions Display Live Chart About Help

F:1 F:2 F:3 F:4 F:5 F:6 F:15 F:16

Tx Count: 8 | Error Count: 0 | Connection Status: Connected

Connection Settings

IP Serial

Protocol TCP IP Address 192.168.178.178 Port 502 Timeout (ms) 3000 Delay Between Polls (ms) 25

Disconnect

Modbus Settings

Function F:15 F:16 Slave Id Register Start Count Poll Rate (ms)

16-Write Multiple Registers 1 8 2 1000

Values | To change the display format, right click on the values

Address	Write Multiple Register
8	0
9	1

16-Write Multiple Registers

Address: [9]

Enter value (-32767 to 32767)

Cancel Send

PowerHud Modbus Tester | Version 3.2.0.1 | 8/8 | Dinko Drazanovic | www.powerhud.com

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